

Data Engineering Lifecycle - Interview Questions & Answers

Q: What is the Data Engineering Lifecycle?

A: It is the process of converting raw data into useful insights through stages like generation, storage, ingestion, transformation, and serving.

Q: Difference between Data Lifecycle and Data Engineering Lifecycle?

A: Data lifecycle covers the entire lifespan of data, while data engineering lifecycle focuses only on stages handled by data engineers.

Q: What are the five stages?

A: Generation, Storage, Ingestion, Transformation, and Serving data.

Q: What are undercurrents?

A: They are foundational aspects like security, DataOps, orchestration, and data management that support all stages.

Q: What is a source system?

A: It is the origin of data such as databases, IoT devices, or applications.

Q: Schemaless vs Fixed schema?

A: Schemaless allows flexible structure, while fixed schema enforces strict structure.

Q: What is data storage?

A: It is where data is stored for processing and access, like data lakes or warehouses.

Q: Hot vs Cold data?

A: Hot data is frequently accessed, cold data is rarely accessed and stored cheaply.

Q: What is data ingestion?

A: Process of collecting data from source systems into storage systems.

Q: Batch vs Streaming?

A: Batch processes data in chunks, streaming processes data in real-time.

Q: Push vs Pull?

A: Push sends data to systems, pull retrieves data from systems.

Q: What is transformation?

A: Process of cleaning, structuring, and preparing data for use.

Q: ETL vs ELT?

A: ETL transforms before loading, ELT loads first then transforms.

Q: What is data serving?

A: Delivering processed data to users for analytics or ML.

Q: What is BI?

A: Business Intelligence analyzes past data for decision making.

Q: What is Reverse ETL?

A: Sending processed data back to operational systems.

Q: Why data quality important?

A: Poor quality leads to incorrect insights and bad decisions.

Q: What is data pipeline?

A: End-to-end flow of data from source to consumption.

Q: What is data lineage?

A: Tracking data flow and transformations across systems.

Q: How to design scalable system?

A: Use distributed systems, proper storage, and efficient pipelines.